QMS-Pharma: Comprehensive Workflow Test Scenarios

Document ID: QMS-TEST-2026-001 **Version:** 1.0 **Date:** 27-Jun-2026 **Prepared By:** QA Engineering Team **Classification:** Internal Use Only

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1. Flowable Workflow Integration Status

#	Module	BPMN Process Key	Backend startProcess	transitionStatus	Frontend Actions	Integration
1	CAPA	capaProcess	Yes (create)	Full switch (9 cases)	getAvailableActions	COMPLETE
2	Deviation	deviationProcess	Yes (create)	Full switch (9 cases)	getAvailableActions	COMPLETE
3	Change Control	changeControlProcess	Yes (create)	Full switch (12 cases)	getAvailableActions	COMPLETE
4	Document	documentProcess	Yes (on transition)	handleWorkflowTransition	Hardcoded buttons	COMPLETE
5	Training	trainingProcess	No	No	Basic start/complete	NOT INTEGRATED
6	Audit	auditProcess	Yes (incomplete vars)	Simple status update	Conditional UI	PARTIAL
7	Complaint	complaintProcess	Yes (create)	Full switch (6 cases)	Conditional UI	COMPLETE
8	Nonconformance	nonconformanceProcess	Yes (create)	Full switch (6 cases)	Conditional UI	COMPLETE
9	Risk	riskProcess	Yes (on transition)	Full switch (8 cases)	Step tracker	MOSTLY COMPLETE
10	Supplier	supplierProcess	Yes (create)	Full switch (8 cases)	Step tracker	COMPLETE
11	Equipment	equipmentCalibrationProcess	No	Partial recordStep	Partial	INCOMPLETE

Summary: 8 of 11 modules are fully wired with Flowable. Training has no integration. Audit and Equipment are partially integrated.

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2. Module 1: CAPA Management

BPMN Process: `capaProcess`

Steps: Initiation -> QA Review -> Root Cause Analysis -> Risk Assessment -> Action Planning -> Action Execution (R3/P7D timer) -> Effectiveness Check (loop-back gateway) -> QA Approval -> Closure

Test Scenario 1.1: Full CAPA Lifecycle (Happy Path)

Step	Action	User Role	API / UI Action	Input Data	Expected Result
1	Create CAPA	End User	POST /api/v1/capas	Title: "OOS Result - Dissolution Test Batch PX-2026-101", Type: BOTH, Priority: HIGH, Source Type: OOS_RESULT, Source Ref: "DEV-2026-001", Dept: Quality Control, Plant: Hyderabad Unit-1, Product: "Paracetamol 500mg Tablets", Batch: "PX-2026-101", Target Date: 30 days from now	Status = INITIATED, CAPA number generated (e.g., CAPA-2026-001), Flowable process started, Workflow step = "Initiation"
2	Submit for Review	Owner	PATCH /api/v1/capas/{id}/status	status: UNDER_REVIEW	Status = UNDER_REVIEW, Workflow step = "QA Review", "Initiation" marked COMPLETED
3	Approve & Start Investigation	QA Reviewer	PATCH /api/v1/capas/{id}/status	status: INVESTIGATION	Status = INVESTIGATION, qaReview Flowable task completed with APPROVED, Workflow step = "Root Cause Analysis"
4	Submit Root Cause Analysis	Owner	POST /api/v1/capas/{id}/root-cause-analysis	method: FIVE_WHY, description: "Investigation into dissolution failure", rootCauses: ["Granulation moisture content exceeded limits"], contributingFactors: ["Humidity control maintenance overdue", "SOP not followed"], fiveWhyEntries: [{level:1, question:"Why did dissolution fail?", answer:"Tablet hardness too high"}, {level:2, question:"Why was hardness high?", answer:"Granulation over-dried"}, {level:3, question:"Why over-dried?", answer:"Moisture sensor incorrect"}, {level:4, question:"Why sensor incorrect?", answer:"Calibration overdue by 2 weeks"}, {level:5, question:"Why calibration overdue?", answer:"PM schedule not tracked"}]	Status = ROOT*CAUSE*IDENTIFIED, RCA saved with 5-Why entries, investigation Flowable task completed

Step	Action	User Role	API / UI Action	Input Data	Expected Result
5	Submit Risk Assessment	Owner	POST /api/v1/capas/{id}/risk-assessment	severity: 4, occurrence: 3, detection: 2, riskLevel: HIGH, justification: "Product quality affected, patient safety risk. Detected during QC testing before release."	Status = ACTION_PLANNING, RPN = 24, riskAssessment Flowable task completed, Workflow step = "Action Planning"
6	Add Corrective Action	Owner	POST /api/v1/capas/{id}/actions	description: "Replace moisture sensor in FBD-03 and recalibrate", type: CORRECTIVE, assignedToId: (Venkat Rao's UUID), dueDate: 14 days from now	Action created with number CAPA-2026-001-A01
7	Add Preventive Action	Owner	POST /api/v1/capas/{id}/actions	description: "Implement automated calibration tracking system with escalation alerts", type: PREVENTIVE, assignedToId: (Anil Prasad's UUID), dueDate: 21 days from now	Action created with number CAPA-2026-001-A02
8	Start Action Execution	Owner	POST /api/v1/capas/{id}/start-action-execution	(empty body)	Status = ACTION*IN*PROGRESS, actionPlanning Flowable task completed, Workflow step = "Action Execution"
9	Complete Corrective Action	Assignee	PATCH /api/v1/capas/{id}/actions/{actionId}/complete	evidence: "Calibration certificate CAL-2026-156 attached"	Action status = COMPLETED, completedDate set
10	Verify Corrective Action	QA Reviewer	PATCH /api/v1/capas/{id}/actions/{actionId}/verify	verificationComments: "Sensor replaced and calibrated. Certificate verified."	Action status = VERIFIED, verifiedBy and verifiedDate set
11	Complete & Verify Preventive Action	Assignee + QA	(Repeat steps 9-10 for second action)	Same pattern	Both actions VERIFIED
12	Complete Action Execution	Owner	POST /api/v1/capas/{id}/complete-action-execution	(empty body)	Status = EFFECTIVENESS_CHECK, actionExecution Flowable task completed
13	Submit Effectiveness Check	QA Reviewer	POST /api/v1/capas/{id}/effectiveness-check	criteria: "No moisture-related OOS for 60 days", checkDate: today, result: EFFECTIVE, evidence: "QC data review - zero OOS for 60+ days", comments: "CAPA effective", requiresRecurrence:	Status = PENDING_CLOSURE, effectivenessCheck task completed with EFFECTIVE

Step	Action	User Role	API / UI Action	Input Data	Expected Result
				true, recurrenceMonths: 6	
14	Close CAPA (E-Sign)	QA Approver	PATCH /api/v1/capas/{id}/status	status: CLOSED, comments: "E-Signed: Closure approved"	Status = CLOSED, closedAt set, qaApproval task completed, Workflow = "Closed"

Test Scenario 1.2: CAPA Rejection at QA Review

Step	Action	User Role	Input Data	Expected Result
1	Create CAPA	End User	Title: "Minor Labeling Typo", Type: CORRECTIVE, Priority: LOW	Status = INITIATED
2	Submit for Review	Owner	status: UNDER_REVIEW	Status = UNDER_REVIEW
3	Reject	QA Reviewer	status: REJECTED, comments: "Not a CAPA-worthy issue. Handle as a deviation."	Status = REJECTED, qaReview task completed with REJECTED, Process ends

Test Scenario 1.3: Effectiveness Check Failure (Loop-back)

Step	Action	User Role	Input Data	Expected Result
1-12	(Follow 1.1 steps 1-12)	Various	(Same as above)	Status = EFFECTIVENESS_CHECK
13	Submit Effectiveness Check - NOT EFFECTIVE	QA Reviewer	result: NOT_EFFECTIVE, evidence: "OOS recurred within 30 days", comments: "Root cause not fully addressed"	Status = ACTION*PLANNING (loop-back), effectivenessCheck task completed with NOT*EFFECTIVE
14	Revise actions and re-execute	Owner	(Add new actions, start execution again)	Cycle through ACTION*IN*PROGRESS -> EFFECTIVENESS_CHECK again

Test Scenario 1.4: Validation - Action Execution Without Actions

Step	Action	User Role	Input Data	Expected Result
1-5	(Follow 1.1 steps 1-5)	Various	(Same as above)	Status = ACTION_PLANNING
6	Attempt Start Execution (no actions)	Owner	POST /start-action-execution	Error 400: "At least one action must be defined before starting execution" (CAPA*NO*ACTIONS)

Test Scenario 1.5: Validation - Complete Execution With Unverified Actions

Step	Action	User Role	Input Data	Expected Result
1-8	(Follow 1.1 steps 1-8)	Various	Status = ACTION*IN*PROGRESS	Actions exist but not verified
9	Attempt Complete Execution	Owner	POST /complete-action-execution	Error 400: "2 actions not yet verified" (CAPA*ACTIONS*NOT_VERIFIED)

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3. Module 2: Deviation Management

BPMN Process: `deviationProcess`

Steps: Reported -> QA Review & Classification -> Investigation (25-day timer) -> Impact Assessment -> Disposition -> CAPA Initiation (conditional) -> Pending Closure -> Closed

Test Scenario 2.1: Full Deviation Lifecycle with CAPA Initiation

Step	Action	User Role	Input Data	Expected Result
1	Create Deviation	End User	Title: "Temperature Excursion in Cold Storage Unit CS-03", Type: UNPLANNED, Category: PROCESS, Dept: Warehouse, Plant: Hyderabad Unit-1, Description: "Temperature recorded at 12 deg C against 2-8 deg C for 45 minutes during power outage", Batch: "INJ-2026-045", Product: "Insulin Glargine Injection"	Status = REPORTED, DEV number generated (DEV-2026-001), Process started
2	Submit for Review	Reporter	PATCH /status, status: UNDER_REVIEW	Status = UNDER_REVIEW, Workflow = "QA Review & Classification"
3	Classify & Assign	QA Reviewer	PATCH /classify, classification: MAJOR, assignedToId: (Investigator UUID), comments: "Potential product impact - requires full investigation"	Status = CLASSIFIED, classification = MAJOR, assignedTo set
4	Start Investigation	Assigned User	PATCH /status, status: INVESTIGATION	Status = INVESTIGATION, Workflow = "Investigation"
5	Submit Investigation	Investigator	PUT /api/v1/deviations/{id}, investigation: "Root cause: UPS battery failure during scheduled power maintenance. Backup generator had 3-minute delay in switchover."	Investigation saved
6	Move to Impact Assessment	Investigator	PATCH /status, status: IMPACT_ASSESSMENT	Status = IMPACT_ASSESSMENT, Validates investigation exists
7	Submit Impact Assessment	QA Reviewer	PUT /api/v1/deviations/{id}, impactAssessment: "Product quality: Insulin stability data shows degradation above 8C for >30 min. 3 batches affected. Potential patient safety impact."	Impact assessment saved
8	Move to Disposition	QA Reviewer	PATCH /status, status: DISPOSITION	Status = DISPOSITION, Validates impact assessment exists
9	Approve Disposition (E-Sign)	QA Approver	PATCH /status, status: CAPA_INITIATED, comments: "E-Signed: Disposition approved. CAPA required for UPS system upgrade."	Status = CAPA_INITIATED, CAPA initiation step recorded
10	Move to Pending Closure	Owner	PATCH /status, status: PENDING_CLOSURE	Status = PENDING_CLOSURE
11	Close Deviation (E-Sign)	QA Approver	PATCH /status, status: CLOSED, comments: "E-Signed: All actions complete."	Status = CLOSED, closedAt set, Process ends

Test Scenario 2.2: Minor Deviation - No CAPA Required

Step	Action	User Role	Input Data	Expected Result
1	Create Deviation	End User	Title: "Incorrect Logbook Entry - Area B2", Type: PLANNED, Category: DOCUMENTATION, Priority: LOW	Status = REPORTED
2-6	Review, Classify (MINOR), Investigate	Various	classification: MINOR	Normal flow
7	Skip CAPA, go to Pending Closure	QA Approver	PATCH /status, status: PENDING_CLOSURE	Status = PENDING_CLOSURE (CAPA step skipped)
8	Close	QA Approver	PATCH /status, status: CLOSED	Status = CLOSED

Test Scenario 2.3: Deviation Rejection

Step	Action	User Role	Input Data	Expected Result
1-2	Create and Submit	End User	Title: "Equipment noise during operation"	Status = UNDER_REVIEW
3	Reject	QA Reviewer	status: REJECTED, comments: "This is a maintenance request, not a quality deviation."	Status = REJECTED, Process ends

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4. Module 3: Change Control

BPMN Process: `changeControlProcess`

Steps: Draft -> Submit -> Impact Assessment (8 dimensions) -> QA Review -> RA Review (conditional) -> Approval (E-Sign) -> Implementation (P30D timer) -> Verification (E-Sign) -> Effectiveness Check (E-Sign) -> Closed

Test Scenario 3.1: Full Change Control with Regulatory Filing

Step	Action	User Role	Input Data	Expected Result
1	Create Change Request	Change Owner	Title: "Upgrade HVAC System in Sterile Manufacturing Area", Type: FACILITY, Category: PRODUCT, Classification: MAJOR, Priority: HIGH, Dept: Engineering, Plant: Hyderabad Unit-2, Description: "Replace existing HVAC with ISO 14644 compliant system for Grade B cleanroom", targetImplementationDate: 90 days from now, regulatoryFilingRequired: true, trainingRequired: true	Status = DRAFT, CHANGE number generated, Flowable process started
2	Submit Change Request	Change Owner	PATCH /status, status: SUBMITTED	Status = SUBMITTED, submitChange task completed, Workflow = "Impact Assessment"
3	Submit Impact Assessment	Change Owner	POST /impact-assessment, productQuality: HIGH, patientSafety: MEDIUM, regulatoryCompliance: HIGH, validationStatus: HIGH, documentation: MEDIUM, training: HIGH, supplierImpact: LOW, stabilityImpact: LOW, overallRiskLevel: HIGH, assessmentSummary: "Major facility change impacting sterile manufacturing. Requires full revalidation."	Impact assessment saved
4	Submit for QA Review	Change Owner	PATCH /status, status: QA_REVIEW	Status = QA_REVIEW, impactAssessment task completed
5	Approve QA Review -> Route to RA Review	QA Reviewer	PATCH /status, status: RA_REVIEW	Status = RA_REVIEW, qaReview completed with regulatoryFilingRequired=true
6	Complete RA Review	RA Reviewer	PATCH /status, status: PENDING_APPROVAL	Status = PENDING_APPROVAL, raReview task completed
7	Add Approvers	QA Reviewer	POST /approvals, userId: (QA Head UUID), approvalOrder: 1	Approver added
8	Approve Change (E-Sign)	QA Approver	PATCH /status, status: APPROVED, comments: "E-Signed: Change approved for implementation"	Status = APPROVED, pendingApproval task completed
9	Start Implementation	Change Owner	PATCH /status, status: IMPLEMENTATION	Status = IMPLEMENTATION
10	Add Implementation Tasks	Change Owner	POST /implementation-tasks, [{description: "Procure HVAC components", assignedToId: UUID, dueDate: +30d}, {description: "Install and commission HVAC", assignedToId: UUID, dueDate: +60d}, {description: "Perform IQ/OQ/PQ validation", assignedToId: UUID, dueDate: +80d}]	3 tasks created
11	Complete All Tasks	Various	PUT /implementation-tasks/{taskId}, status: COMPLETED	All tasks = COMPLETED
12	Complete Implementation	Change Owner	PATCH /status, status: VERIFICATION	Status = VERIFICATION, actualImplementationDate set
13	Verify (E-Sign)	QA Reviewer	PATCH /status, status: EFFECTIVENESS_CHECK, comments: "E-Signed: Implementation verified"	Status = EFFECTIVENESS_CHECK
14	Close (E-Sign)	QA Approver	PATCH /status, status: CLOSED, comments: "E-Signed: Change effective and closed"	Status = CLOSED, closedDate set

Test Scenario 3.2: Change Control Without Regulatory Filing

Step	Action	User Role	Input Data	Expected Result
1	Create	Change Owner	Title: "Update SOP-QC-012 Dissolution Testing", Type: DOCUMENT, Classification: MINOR, regulatoryFilingRequired: false	Status = DRAFT
2-4	Submit, Impact, QA Review	Various	(Normal flow)	Status = QA_REVIEW
5	Approve QA Review -> Skip RA	QA Reviewer	PATCH /status, status: PENDING_APPROVAL	Status = PENDING*APPROVAL (RA*REVIEW skipped), qaReview completed with regulatoryFilingRequired=false
6-9	Approve, Implement, Verify, Close	Various	(Normal flow)	Status = CLOSED

Test Scenario 3.3: Change Rejection at Approval

Step	Action	User Role	Input Data	Expected Result
1-6	(Follow 3.1 steps 1-6)	Various	(Same as above)	Status = PENDING_APPROVAL
7	Reject	QA Approver	status: REJECTED, comments: "Cost-benefit analysis insufficient. Resubmit with updated financial projections."	Status = REJECTED, Process ends

Test Scenario 3.4: Reopen Implementation from Effectiveness Check

Step	Action	User Role	Input Data	Expected Result
1-13	(Follow 3.1 steps 1-13)	Various	(Same)	Status = EFFECTIVENESS_CHECK
14	Reopen Implementation	QA Reviewer	status: IMPLEMENTATION, comments: "Particle count in Zone B2 exceeds limits. Additional balancing required."	Status = IMPLEMENTATION (loop-back), Effectiveness Check marked completed with comment

Test Scenario 3.5: Validation - QA Review Without Impact Assessment

Step	Action	User Role	Input Data	Expected Result
1-2	Create and Submit	Change Owner	(Normal)	Status = SUBMITTED
3	Attempt QA Review (no impact)	Change Owner	PATCH /status, status: QA_REVIEW	Error 400: "Impact assessment must be submitted before QA review" (CC*NO*IMPACT_ASSESSMENT)

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5. Module 4: Document Management

BPMN Process: `documentProcess`

Steps: Draft -> Draft Review (APPROVED/REVISION_REQUIRED loop) -> QA Approval -> Training Assignment -> Effective -> Periodic Review (P365D timer)

Test Scenario 4.1: Full Document Lifecycle

Step	Action	User Role	Input Data	Expected Result
1	Create Document	Author	Title: "SOP-PRD-045 Granulation Process Control", Type: SOP, Category: PRODUCTION, Version: "1.0", Description: "Standard operating procedure for wet granulation process control including in-process checks", Dept: Production, Plant: Hyderabad Unit-1, effectiveDate: 14 days from now	Status = DRAFT, Document number generated
2	Upload Content	Author	POST /upload, file: SOP-PRD-045-v1.0.pdf	Document file stored in GCS
3	Submit for Review	Author	PATCH /status, status: PENDING_REVIEW	Status = PENDING_REVIEW, Flowable process started, Workflow = "Draft Review"
4	Approve Review	Reviewer	PATCH /status, status: PENDING_APPROVAL, comments: "Content reviewed and technically accurate"	Status = PENDING_APPROVAL, draftReview task completed with APPROVED
5	Approve Document	QA Approver	PATCH /status, status: APPROVED, comments: "Approved for training and release"	Status = APPROVED, qaApproval task completed
6	Make Effective	System/Admin	PATCH /status, status: EFFECTIVE	Status = EFFECTIVE, effectiveDate set

Test Scenario 4.2: Document Review - Revision Required

Step	Action	User Role	Input Data	Expected Result
1-3	Create, Upload, Submit	Author	(Same as 4.1)	Status = PENDING_REVIEW
4	Request Revision	Reviewer	PATCH /status, status: DRAFT, comments: "Section 4.3 missing in-process moisture check parameters. Please revise."	Status = DRAFT (loop-back), Author revises
5	Resubmit	Author	PATCH /status, status: PENDING_REVIEW	Status = PENDING_REVIEW (re-enters review cycle)
6	Approve	Reviewer	PATCH /status, status: PENDING_APPROVAL	Status = PENDING_APPROVAL

Test Scenario 4.3: Document Version Upgrade

Step	Action	User Role	Input Data	Expected Result
1	Create new version of existing doc	Author	Title: "SOP-PRD-045 Granulation Process Control", Version: "2.0", parentDocumentId: (v1.0 UUID), changeDescription: "Added moisture sensor verification step per CAPA-2026-001"	New version created, old version status unchanged until new is effective
2-6	(Follow 4.1 steps 2-6)	Various	(Same flow)	v2.0 becomes EFFECTIVE, v1.0 becomes SUPERSEDED

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6. Module 5: Training Management

BPMN Process: `trainingProcess`

Steps: Complete Training -> Assessment (conditional) -> Trainer Verification (loop-back if not effective) -> Record Completion
Note: Training module currently has NO Flowable integration. Status transitions are database-driven only.

Test Scenario 5.1: Training Curriculum and Assignment

Step	Action	User Role	Input Data	Expected Result
1	Create Curriculum	Training Coordinator	Title: "Aseptic Gowning Procedure v2.0", Type: SOP_TRAINING, Description: "Updated gowning technique training per CAPA-2026-003", documentId: (SOP UUID), passingScore: 80, validityMonths: 12, assessmentRequired: true	Curriculum created
2	Create Assignment	Training Coordinator	curriculumId: (above), traineeId: (Operator UUID), trainerId: (Trainer UUID), dueDate: 14 days from now	Assignment created, status = ASSIGNED
3	Start Training	Trainee	PATCH /assignments/{id}/start	Status = IN_PROGRESS
4	Complete Training	Trainee	PATCH /assignments/{id}/complete, score: 92, comments: "Completed video-based assessment and practical demonstration"	Status = COMPLETED, score recorded

Test Scenario 5.2: Training - Failed Assessment

Step	Action	User Role	Input Data	Expected Result
1-3	(Follow 5.1 steps 1-3)	Various	(Same)	Status = IN_PROGRESS
4	Complete with failing score	Trainee	score: 65 (below passing score of 80)	Status = COMPLETED but score < passingScore, requires reassignment

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7. Module 6: Audit Management

BPMN Process: `auditProcess`

Steps: Audit Planning -> Plan Approval -> Audit Execution (dueDate timer) -> Findings Review -> CAPA Gateway -> Auditee Response -> Audit Closure **Note:** Audit module has partial Flowable integration. Process starts but tasks are not completed via workflow engine.

Test Scenario 6.1: Full Internal Audit Lifecycle

Step	Action	User Role	Input Data	Expected Result
1	Create Audit	QA Manager	Title: "Annual GMP Compliance Audit - Production Area", Type: INTERNAL, scope: "Production processes, documentation, personnel training, facility maintenance", leadAuditorId: (Auditor UUID), auditeeId: (Production Head UUID), Dept: Production, Plant: Hyderabad Unit-1, scheduledStartDate: 14 days from now, scheduledEndDate: 16 days from now	Status = PLANNED, Audit number generated, Flowable process started
2	Schedule Audit	Lead Auditor	PATCH /status, status: SCHEDULED	Status = SCHEDULED
3	Start Audit	Lead Auditor	PATCH /status, status: IN_PROGRESS	Status = IN_PROGRESS, actualStartDate set
4	Add Finding - Critical	Lead Auditor	POST /findings, title: "Inadequate Temperature Monitoring in Warehouse Zone C", type: CRITICAL, description: "No continuous temperature monitoring installed. Manual checks only twice daily. Risk of excursion going undetected.", clause: "WHO TRS 986 Annex 9, Section 8.4", area: "Warehouse Zone C"	Finding created
5	Add Finding - Major	Lead Auditor	POST /findings, title: "Training Records Incomplete for 3 Operators", type: MAJOR, description: "Operators OP-12, OP-15, OP-22 missing signed training records for SOP-PRD-045", clause: "Schedule M, Part I, Section 12.4"	Finding created
6	Add Finding - Observation	Lead Auditor	POST /findings, title: "Equipment ID labels faded in Area B1", type: OBSERVATION, description: "Equipment identification labels on 4 units are faded and difficult to read"	Finding created
7	Draft Report	Lead Auditor	PATCH /status, status: REPORT*DRAFTING, auditReport: {summary: "Overall GMP compliance satisfactory with 1 critical and 1 major finding requiring immediate attention", overallRating: "SATISFACTORY*WITH_FINDINGS"}	Status = REPORT_DRAFTING
8	Submit for Review	Lead Auditor	PATCH /status, status: UNDER_REVIEW	Status = UNDER_REVIEW
9	Auditee Response	Auditee	PATCH /findings/{findingId}/respond, response: "Continuous monitoring system procurement initiated. Expected installation within 30 days."	Finding response recorded
10	Initiate CAPA for Critical Finding	QA Manager	POST /findings/{findingId}/initiate-capa	CAPA record created, linked to audit finding
11	Complete Audit	QA Approver	PATCH /status, status: COMPLETED	Status = COMPLETED, actualEndDate set

Test Scenario 6.2: Supplier Audit

Step	Action	User Role	Input Data	Expected Result
1	Create Audit	QA Manager	Title: "API Supplier Qualification Audit - PharmaChem Ltd", Type: SUPPLIER, supplierId: (Supplier UUID), scope: "GMP compliance, quality systems, documentation", scheduledStartDate: 30 days from now	Status = PLANNED
2-11	(Follow similar flow as 6.1)	Various	(Supplier-specific findings)	Full audit cycle for supplier

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8. Module 7: Complaint Management

BPMN Process: `complaintProcess`

Steps: Received -> Initial Assessment & Classification -> Investigation (conditional, 20-day timer) -> Disposition Review -> CAPA (conditional) -> Response Preparation -> Final Review & Closure

Test Scenario 7.1: Product Complaint with Investigation and CAPA

Step	Action	User Role	Input Data	Expected Result
1	Create Complaint	End User	Title: "Tablet Discoloration - Batch MF-2026-078", Source: CUSTOMER, Type: PRODUCT_QUALITY, Priority: HIGH, Description: "Customer reports yellow-brown spots on Metformin 500mg tablets from batch MF-2026-078. 3 bottles affected.", productName: "Metformin 500mg Tablets", batchNumber: "MF-2026-078", complaintDate: today, customerName: "City Pharmacy, Hyderabad", contactEmail: "pharmacy@example.com"	Status = RECEIVED, Complaint number generated, Flowable process started
2	Initial Assessment	QA Reviewer	PATCH /status + assessment data, classification: MAJOR, investigationRequired: true, adverseEventReported: false, initialAssessment: "Potential cross-contamination or degradation. Retain samples collected. Full investigation required."	Status = CLASSIFIED, routed to investigation
3	Complete Investigation	Investigator	PATCH /status, status: INVESTIGATION_COMPLETE, investigation: "Root cause: Iron contamination from worn FBD screen mesh. Particle analysis confirms Fe_2O_3 deposits. Screen replaced and validated."	Status = INVESTIGATION_COMPLETE
4	Disposition Review	QA Approver	PATCH /status, status: RESPONSE_PENDING, disposition: "Batch recall limited to affected distribution. Replacement product dispatched. CAPA required for equipment maintenance program."	Status = RESPONSE_PENDING
5	Prepare & Send Response	QA Reviewer	PATCH /status, status: RESPONSE_SENT, responseText: "Dear Customer, thank you for reporting this quality concern. Investigation confirmed an isolated manufacturing issue which has been corrected. Replacement product has been shipped. Reference: COMPLAINT-2026-001"	Status = RESPONSE_SENT
6	Close Complaint	QA Approver	PATCH /status, status: CLOSED	Status = CLOSED

Test Scenario 7.2: Complaint - No Investigation Required

Step	Action	User Role	Input Data	Expected Result
1	Create Complaint	End User	Title: "Packaging Damage During Shipping", Source: DISTRIBUTOR, Type: PACKAGING, Priority: LOW	Status = RECEIVED
2	Assess - No Investigation	QA Reviewer	classification: MINOR, investigationRequired: false	Status = CLASSIFIED, skips investigation
3	Prepare Response	QA Reviewer	PATCH /status, status: RESPONSE_SENT	Status = RESPONSE_SENT
4	Close	QA Approver	PATCH /status, status: CLOSED	Status = CLOSED

Test Scenario 7.3: Adverse Event Complaint

Step	Action	User Role	Input Data	Expected Result
1	Create Complaint	End User	Title: "Patient Allergic Reaction Report", Source: HEALTHCARE*PROVIDER, Type: ADVERSE*EVENT, Priority: CRITICAL	Status = RECEIVED
2	Assess - Adverse Event	QA Reviewer	classification: CRITICAL, investigationRequired: true, adverseEventReported: true	Status = CLASSIFIED, triggers regulatory reporting path
3-6	Investigation through Closure	Various	(Full investigation with regulatory reporting documentation)	Complete lifecycle with regulatory notification

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9. Module 8: Nonconformance Management

BPMN Process: `nonconformanceProcess`

Steps: Identified -> Hold (conditional) -> Initial Review & Classification -> Investigation -> Disposition Review (6 types) -> CAPA (conditional) -> Final Review & Closure -> Release Hold

Test Scenario 8.1: NC with Material Hold and CAPA

Step	Action	User Role	Input Data	Expected Result
1	Create NC	End User	Title: "API Assay Failure - Metformin Batch API-2026-034", Type: MATERIAL, Description: "Incoming QC testing shows assay at 97.2% against specification of 98.0-102.0%", productName: "Metformin HCl API", batchNumber: "API-2026-034", Dept: Quality Control, Plant: Hyderabad Unit-1, stageDetected: INCOMING, supplierId: (Supplier UUID)	Status = IDENTIFIED, NC number generated, Process started
2	Place Material on Hold	QA Reviewer	PATCH /hold, action: HOLD	Material hold placed, holdDate set
3	Review & Classify	QA Reviewer	PATCH /status, status: UNDER_REVIEW, classification: CRITICAL	Status = UNDER_REVIEW, reviewDecision=VALID
4	Complete Investigation	Investigator	PATCH /status, status: INVESTIGATION_COMPLETE, findings: "Supplier batch record review shows deviation in crystallization temperature. CoA values marginally passing at supplier site."	Status = INVESTIGATION_COMPLETE
5	Submit Disposition	QA Approver	PATCH /disposition, dispositionDecision: RETURN*TO*SUPPLIER, justification: "Material does not meet incoming specifications. Return to supplier with NCR.", capaRequired: true	Status = DISPOSITION, routes to CAPA_PENDING
6	Complete CAPA Initiation	QA Reviewer	PATCH /status, status: PENDING_CLOSURE	Status = PENDING_CLOSURE
7	Close NC	QA Approver	PATCH /status, status: CLOSED	Status = CLOSED
8	Release Hold	QA Reviewer	PATCH /hold, action: RELEASE	Material hold released (material returned to supplier)

Test Scenario 8.2: NC Voided

Step	Action	User Role	Input Data	Expected Result
1	Create NC	End User	Title: "Suspected Particle in Vial - Lot VL-2026-001"	Status = IDENTIFIED
2	Void NC	QA Reviewer	PATCH /status, status: VOID, comments: "Upon re-examination, particle confirmed as air bubble. Not a valid NC."	Status = VOID, reviewDecision=VOID, Process ends

Test Scenario 8.3: NC Disposition - Use As Is

Step	Action	User Role	Input Data	Expected Result
1-4	Create through Investigation	Various	Type: PROCESS, stageDetected: IN_PROCESS	Status = INVESTIGATION_COMPLETE
5	Disposition: Use As Is	QA Approver	dispositionDecision: USE*AS*IS, justification: "Deviation within acceptable limits per ICH Q7 Section 14.3. Risk assessment confirms no impact on product quality."	Status = PENDING_CLOSURE (no CAPA needed)
6	Close	QA Approver	PATCH /status, status: CLOSED	Status = CLOSED

---

10. Module 9: Risk Management

BPMN Process: `riskProcess`

Steps: Risk Register Created -> Risk Evaluation (FMEA/HACCP) -> Control Planning -> Control Implementation -> Residual Risk Assessment (loop-back if unacceptable) -> Register Approval (E-Sign) -> Periodic Review (timer)

Test Scenario 9.1: Full Risk Register Lifecycle

Step	Action	User Role	Input Data	Expected Result
1	Create Risk Register	Risk Owner	Title: "Process Risk Assessment - Tablet Compression", riskType: PROCESS, Description: "FMEA analysis for tablet compression process parameters", methodology: FMEA, Dept: Production, Plant: Hyderabad Unit-1	Register created, status = DRAFT
2	Submit for Assessment	Risk Owner	PATCH /registers/{id}/status, status: IN_ASSESSMENT	Status = IN_ASSESSMENT, Flowable process started
3	Add Risk Assessment Item	QA Reviewer	POST /registers/{id}/assessments, hazard: "Tablet weight variation", failureMode: "Compression force inconsistency", severity: 4, occurrence: 3, detection: 2, riskLevel: HIGH, currentControls: "In-process weight checks every 30 min"	Assessment item created, RPN = 24
4	Add Another Assessment	QA Reviewer	POST /assessments, hazard: "Tablet hardness OOS", failureMode: "Granulation moisture variation", severity: 3, occurrence: 2, detection: 3, riskLevel: MEDIUM	Assessment item created, RPN = 18
5	Complete Evaluation	QA Reviewer	PATCH /status, status: EVALUATION	Status = EVALUATION
6	Add Risk Controls	Risk Owner	POST /registers/{id}/controls, [{description: "Install real-time compression force monitoring", type: PREVENTIVE, assignedToId: UUID}, {description: "Automated weight check with auto-reject", type: DETECTIVE, assignedToId: UUID}]	Controls created
7	Complete Control Planning	Risk Owner	PATCH /status, status: CONTROL_IMPLEMENTATION	Status = CONTROL_IMPLEMENTATION
8	Complete Implementation	Risk Owner	PATCH /status, status: RESIDUAL*RISK*REVIEW	Status = RESIDUAL*RISK*REVIEW
9	Assess Residual Risk	QA Reviewer	PATCH /assessments/{id}/residual-risk, residualSeverity: 4, residualOccurrence: 1, residualDetection: 1, residualRiskLevel: LOW	Residual RPN = 4, risk significantly reduced
10	Accept Residual Risk	QA Approver	PATCH /status, status: PENDING_APPROVAL	Status = PENDING_APPROVAL, residualDecision=ACCEPTABLE
11	Approve Register (E-Sign)	QA Approver	PATCH /status, status: APPROVED, comments: "E-Signed: Risk register approved"	Status = APPROVED, approvedBy set, nextReviewDate calculated

Test Scenario 9.2: Unacceptable Residual Risk (Loop-back)

Step	Action	User Role	Input Data	Expected Result
1-8	(Follow 9.1 steps 1-8)	Various	(Same)	Status = RESIDUAL*RISK*REVIEW
9	Assess Residual Risk - Still High	QA Reviewer	residualSeverity: 4, residualOccurrence: 3, residualDetection: 2, residualRiskLevel: HIGH	Residual RPN still 24
10	Reject Residual Risk	QA Approver	PATCH /status, status: BACK*TO*CONTROL, comments: "Additional controls needed"	Status loops back to CONTROL_IMPLEMENTATION
11	Add more controls and re-assess	Risk Owner	(Add controls, implement, re-assess)	Cycle until residual risk acceptable

11. Module 10: Supplier Quality Management

BPMN Process: `supplierProcess`

Steps: Pending -> Document Review -> Supplier Audit (conditional) -> Corrective Action (if audit failed) -> Qualification Approval -> Qualified -> Performance Monitoring (ongoing)

Test Scenario 10.1: Full Supplier Qualification with Audit

Step	Action	User Role	Input Data	Expected Result
1	Create Supplier	Procurement	Name: "PharmaChem Industries Pvt Ltd", supplierType: API, category: CRITICAL, contactPerson: "Dr. Suresh Kumar", email: "quality@pharmachem.com", phone: "+91-40-23456789", address: "Plot 42, APIIC Industrial Area, Hyderabad", country: "India", qualificationOwnerId: (QA Manager UUID), certifications: ["WHO-GMP", "ISO 9001:2015"]	Status = PENDING_QUALIFICATION, Supplier number generated, Process started, Workflow = "Document Review"
2	Document Review - Audit Required	QA Reviewer	PATCH /status, status: UNDER*EVALUATION, documentDecision: AUDIT*REQUIRED	Status = UNDER_EVALUATION, routes to audit task
3	Conduct Audit & Pass	Auditor	PATCH /status with auditResult: PASSED	Status progresses, audit marked passed
4	Approve Qualification	QA Approver	PATCH /status, status: QUALIFIED	Status = QUALIFIED, approvalDecision=APPROVED, qualificationDate set
5	Update Performance Scores	QA Reviewer	PATCH /scores, qualityScore: 85, deliveryScore: 90, documentationScore: 80, complianceScore: 88	Scores updated, overall score calculated

Test Scenario 10.2: Supplier Audit Failure with Corrective Action

Step	Action	User Role	Input Data	Expected Result
1-2	Create and Review	Various	(Same as 10.1)	Status = UNDER_EVALUATION, audit required
3	Audit Failed	Auditor	PATCH /status with auditResult: FAILED	Status = CORRECTIVE*ACTION*REQUIRED
4	Complete Corrective Actions	Supplier/QA	PATCH /status, status: PENDING_APPROVAL	Status = PENDING_APPROVAL, corrective actions recorded
5	Approve	QA Approver	PATCH /status, status: QUALIFIED	Status = QUALIFIED

Test Scenario 10.3: Supplier Disqualification from Score Degradation

Step	Action	User Role	Input Data	Expected Result
1	(Existing qualified supplier)	-	Status = QUALIFIED	-
2	Update Scores - Poor Performance	QA Reviewer	qualityScore: 45, deliveryScore: 50, documentationScore: 40	Overall score drops below threshold
3	Place on Probation	QA Manager	PATCH /status, status: ON_PROBATION	Status = ON_PROBATION
4	No improvement -> Disqualify	QA Manager	PATCH /status, status: DISQUALIFIED	Status = DISQUALIFIED

Test Scenario 10.4: Supplier Rejection at Document Review

Step	Action	User Role	Input Data	Expected Result
1	Create Supplier	Procurement	(Same as 10.1)	Status = PENDING_QUALIFICATION
2	Document Review - Reject	QA Reviewer	PATCH /status, status: REJECTED, comments: "Missing WHO-GMP certificate. Insufficient quality documentation."	Status = REJECTED, documentDecision=REJECTED

---

12. Module 11: Equipment Management

BPMN Process: `equipmentCalibrationProcess`

Steps: Registration -> IQ -> OQ -> PQ -> Operational -> Calibration Cycle (timer) -> Calibration Execution -> Calibration Review -> Maintenance (timer) -> Re-qualification (P365D) -> Decommission **Note:** Equipment module has partial Flowable integration. Qualification and calibration use recordStep but not BPMN task completion.

Test Scenario 11.1: Equipment Registration and Qualification

Step	Action	User Role	Input Data	Expected Result
1	Create Equipment	Engineer	Name: "Fluid Bed Dryer FBD-04", equipmentType: MANUFACTURING, manufacturer: "GEA Process Engineering", model: "FlexStream Multi-Processor 5", serialNumber: "GEA-FBD-2026-0042", location: "Production Block A, Room 201", Dept: Production, Plant: Hyderabad Unit-1, gxpRelevant: true, calibrationRequired: true, calibrationFrequencyDays: 180	Status = ACTIVE, qualificationStatus = NOT_QUALIFIED, Equipment number generated
2	Complete IQ	Engineer	POST /equipment/{id}/qualify/IQ	qualificationStatus updated, "IQ Completed" step recorded
3	Complete OQ	Engineer	POST /equipment/{id}/qualify/OQ	"OQ Completed" step recorded
4	Complete PQ	Engineer	POST /equipment/{id}/qualify/PQ	qualificationStatus = QUALIFIED, "PQ Completed" + "Equipment Operational" steps recorded

Test Scenario 11.2: Equipment Calibration Cycle

Step	Action	User Role	Input Data	Expected Result
1	Create Calibration Record	Technician	POST /equipment/{id}/calibrations, calibrationType: ROUTINE, calibrationDate: today, performedById: (Technician UUID), instrumentUsed: "Fluke 1524 Reference Thermometer", certificateNumber: "CAL-2026-0089"	Calibration record created
2	Record Result - PASS	Technician	PATCH /calibrations/{id}, result: PASS, readings: "Range: 20-200 deg C, Accuracy: +/- 0.1 deg C within specification", nextCalibrationDate: +180 days	calibrationStatus = CALIBRATED, nextCalibrationDate set

Test Scenario 11.3: Calibration Failure

Step	Action	User Role	Input Data	Expected Result
1	Create Calibration	Technician	(Same as 11.2)	Calibration created
2	Record Result - FAIL	Technician	result: FAIL, readings: "Temperature offset: +2.3 deg C at 100 deg C reference point. Outside tolerance of +/- 0.5 deg C"	calibrationStatus = OUT*OF*CALIBRATION, equipment may need OUT*OF*SERVICE status
3	Equipment Out of Service	QA Reviewer	PATCH /equipment/{id}, status: OUT*OF*SERVICE	Status = OUT*OF*SERVICE until repaired and re-calibrated

Test Scenario 11.4: Preventive Maintenance

Step	Action	User Role	Input Data	Expected Result
1	Create Maintenance Record	Engineer	POST /equipment/{id}/maintenance, type: PREVENTIVE, description: "6-monthly PM - filter replacement, belt inspection, bearing lubrication", scheduledDate: today, assignedToId: (Technician UUID)	Maintenance record created
2	Complete Maintenance	Technician	PATCH /maintenance/{id}/complete, completionNotes: "All PM tasks completed per checklist. New HEPA filter installed (lot HF-2026-012). Belt tension adjusted.", completedDate: today	Maintenance completed, nextMaintenanceDate calculated

Test Scenario 11.5: Equipment Decommission

Step	Action	User Role	Input Data	Expected Result
1	Decommission Equipment	QA Manager	POST /equipment/{id}/decommission, reason: "End of useful life. Replacement unit FBD-05 installed and qualified.", disposalMethod: "Vendor buyback program"	Status = DECOMMISSIONED, "Decommissioned" workflow step recorded

---

13. Integrated Cross-Module Test Scenarios

Scenario 13.1: Deviation -> CAPA -> Change Control (Full Chain)

This tests the most common cross-module workflow in pharmaceutical QMS.

Step	Module	Action	Input Data	Expected Result
1	Deviation	Create DEV-2026-010	Title: "Repeated OOS in Dissolution - Compression Machine CM-02", Category: EQUIPMENT	DEV created
2	Deviation	Full workflow through CAPA_INITIATED	Investigation finds worn punch tooling as root cause	DEV at CAPA_INITIATED status
3	CAPA	Create CAPA-2026-010	Title: "Compression Tooling Management Program", sourceType: DEVIATION, sourceReference: "DEV-2026-010"	CAPA created, linked to deviation
4	CAPA	Full workflow through ACTION_PLANNING	RCA: 5-Why confirms tooling wear tracking inadequate. Risk Assessment: Severity=4, Occurrence=3, Detection=2	CAPA at ACTION_PLANNING
5	CAPA	Add preventive action	"Implement preventive tooling replacement program with usage-based tracking"	Action added
6	Change Control	Create CHANGE-2026-005	Title: "Implement Tooling Lifecycle Management System", sourceReference: "CAPA-2026-010", Type: PROCESS, Classification: MAJOR	CC created, linked to CAPA
7	Change Control	Full workflow through CLOSED	Impact assessment, approvals, implementation, verification	CC closed
8	CAPA	Complete action execution, effectiveness check	All actions verified, effectiveness confirmed	CAPA at PENDING_CLOSURE
9	CAPA	Close CAPA	E-signature closure	CAPA closed
10	Deviation	Close deviation	E-signature closure	DEV closed
Verify	All	Audit trail check	Query audit trails for all 3 records	Complete chain of linked records visible

Scenario 13.2: Complaint -> NC -> CAPA -> Supplier Action

Step	Module	Action	Input Data	Expected Result
1	Complaint	Create COMPLAINT-2026-005	Title: "Foreign Particle in API Batch", Source: CUSTOMER, Type: PRODUCT_QUALITY, Priority: CRITICAL	Complaint created
2	Complaint	Classify & Investigate	classification: CRITICAL, investigationRequired: true	Investigation initiated
3	NC	Create NC-2026-008	Title: "Foreign Particle Contamination - API Batch SUP-2026-089", Type: MATERIAL, stageDetected: INCOMING, supplierId: (Supplier UUID), sourceReference: "COMPLAINT-2026-005"	NC created, linked to complaint
4	NC	Full workflow - Disposition: Return to Supplier	dispositionDecision: RETURN*TO*SUPPLIER, capaRequired: true	NC at CAPA_PENDING
5	CAPA	Create CAPA-2026-012	Title: "Supplier Quality Improvement - Foreign Particle Prevention", sourceType: NONCONFORMANCE, sourceReference: "NC-2026-008"	CAPA created
6	Supplier	Update supplier scores	qualityScore: 55 (downgrade)	Supplier score updated
7	Supplier	Place on probation	PATCH /status, status: ON_PROBATION	Supplier on probation
8	CAPA	Full lifecycle through closure	Actions include supplier audit, enhanced incoming inspection	CAPA closed
9	NC	Close NC	All dispositions complete	NC closed
10	Complaint	Close complaint	Response sent to customer	Complaint closed
Verify	All	Cross-reference check	All records reference each other	Full traceability from complaint to supplier action

Scenario 13.3: Audit -> Finding -> CAPA -> Training -> Document Update

Step	Module	Action	Input Data	Expected Result
1	Audit	Create Internal Audit	Title: "Annual Schedule M Compliance - Documentation", Type: INTERNAL	Audit created
2	Audit	Add Critical Finding	"SOP-PRD-045 not updated for 3 years. Current practices diverge from documented procedure."	Finding recorded
3	Audit	Initiate CAPA from Finding	POST /findings/{id}/initiate-capa	CAPA created, linked to audit finding
4	CAPA	Investigate & Plan Actions	Corrective: "Update SOP-PRD-045 to reflect current practices"	Actions planned
5	Document	Create new version of SOP-PRD-045	Title: "SOP-PRD-045 v3.0", changeDescription: "Updated per CAPA-2026-015 and Audit Finding"	Document v3.0 created
6	Document	Full review and approval cycle	Review, QA approval	Document EFFECTIVE
7	Training	Create curriculum for updated SOP	Title: "SOP-PRD-045 v3.0 Training"	Curriculum created
8	Training	Assign to all affected personnel	Create assignments for 15 operators	15 assignments created
9	Training	All complete training	Each trainee completes with passing score	All assignments COMPLETED
10	CAPA	Verify effectiveness	"All operators trained on updated SOP. No deviations in 60 days."	CAPA closed
11	Audit	Close audit	All findings addressed	Audit COMPLETED

Scenario 13.4: Equipment Calibration Failure -> Deviation -> Risk Update

Step	Module	Action	Input Data	Expected Result
1	Equipment	Calibration fails for FBD-03 moisture sensor	result: FAIL, "Temperature offset exceeds tolerance"	Equipment OUT*OF*CALIBRATION
2	Equipment	Set OUT*OF*SERVICE	status: OUT*OF*SERVICE	Equipment cannot be used
3	Deviation	Create DEV-2026-015	Title: "FBD-03 Moisture Sensor Calibration Failure", Type: UNPLANNED, Category: EQUIPMENT	Deviation created
4	Deviation	Investigate - Identify affected batches	"3 batches processed since last valid calibration. Batches: PX-2026-110, PX-2026-111, PX-2026-112"	Investigation complete
5	Risk	Update risk register for tablet compression	Add new assessment item for sensor failure mode	Risk register updated
6	Equipment	Repair and re-calibrate	result: PASS after sensor replacement	Equipment back to CALIBRATED
7	Equipment	Set ACTIVE	status: ACTIVE	Equipment back in service
8	Deviation	Close deviation	All affected batches evaluated, CAPA if needed	Deviation closed

Scenario 13.5: Change Control -> Document -> Training (Sequential Dependencies)

Step	Module	Action	Input Data	Expected Result
1	Change Control	Create CC for new cleaning validation protocol	Title: "Implement Dedicated Cleaning Validation Protocol for Multi-Product Facility", trainingRequired: true	CC created
2	Change Control	Impact Assessment	documentation: HIGH, training: HIGH, validationStatus: HIGH	High documentation/training impact
3	Change Control	Full approval cycle	E-signature approvals	CC APPROVED
4	Document	Create new SOP	Title: "SOP-CLN-001 Cleaning Validation Protocol"	Document created
5	Document	Review and approve	Full document lifecycle	Document EFFECTIVE
6	Training	Create training curriculum	Linked to new SOP-CLN-001	Curriculum created
7	Training	Assign and complete for all production staff	20 assignments	All completed
8	Change Control	Complete implementation	All tasks done: SOP created, training complete	Implementation complete
9	Change Control	Verification and Closure	E-signature verification and closure	CC CLOSED

---

14. Sample User Accounts for Testing

User Credentials and Roles

User Name	Role Codes	Department	Purpose
Rajesh Kumar	OWNER, END_USER	Quality Control	CAPA Owner, Deviation Reporter
Srinivas Rao	QA_REVIEWER	Quality Assurance	QA Review, Classification
Priya Sharma	QA*REVIEWER, QA*APPROVER	Quality Assurance	QA Review & Approval, E-Sign
Suresh Reddy	OWNER, END_USER	Production	Production Owner, Deviation Assignee
Anitha Rao	END_USER	Quality Control	End User, Reporter
Lakshmi Devi	QA*APPROVER, VAULT*ADMIN	Quality Assurance	Final Approvals, Admin
Venkat Rao	END_USER, REVIEWER	Engineering	Engineering tasks, Action Assignee
Mohammad Ali	END_USER	Warehouse	Warehouse operations
Kavitha Reddy	QA_REVIEWER	Quality Assurance	QA Review, Complaint Assessment
Ravi Teja	OWNER	Production	Training coordination, Production tasks
Deepa Menon	REVIEWER, AUDITOR	Quality Assurance	Audit lead, Document Review
Ramesh Gupta	APPROVER	Management	Senior Management Approvals

API Authentication

All API calls require Bearer token authentication:

```
Authorization: Bearer <access_token>
```

Login: POST /api/v1/auth/login with { username, password }

Test Database

- PostgreSQL on port 5434, database: qms
- Backend API on port 8082
- Shell App on port 4200
- CAPA MFE on port 4201
- Deviation MFE on port 4202
- Change Control MFE on port 4203
- Document MFE on port 4204
- Training MFE on port 4205
- QMS Core MFE on port 4206 (Audit, Risk, Supplier, Complaint, NC, Equipment)

Appendix A: Workflow Verification Queries

Check Flowable Process Definitions

```
SELECT key_, name_, version_, deployment_id_
  FROM act_re_procdef
  ORDER BY key_;
```

Expected: 11 process definitions (capaProcess, deviationProcess, changeControlProcess, documentProcess, trainingProcess, auditProcess, complaintProcess, nonconformanceProcess, riskProcess, supplierProcess, equipmentCalibrationProcess)

Check Active Tasks for a Record

```
SELECT t.id_, t.name_, t.task_def_key_, t.assignee_, t.create_time_
  FROM act_ru_task t
  JOIN act_ru_execution e ON t.execution_id_ = e.id_
  WHERE e.proc_inst_id_ = '<flowable_process_id>';
```

Check Workflow History for a Record

```
SELECT step_name, status, assigned_to_id, started_at, completed_at, comments, step_order
  FROM workflow_history
  WHERE record_type = 'CAPA' AND record_id = '<capa_uuid>'
  ORDER BY step_order;
```

Check Audit Trail for a Record

```
SELECT action, field_name, old_value, new_value, reason_for_change, timestamp, user_id
  FROM audit_trail
  WHERE record_type = 'CAPA' AND record_id = '<capa_uuid>'
  ORDER BY timestamp;
```

Task Inbox Query

```
GET /api/v1/tasks/inbox
```

Returns all tasks assigned to or claimable by the current user across all modules.

Appendix B: Key Validation Rules Summary

Module	Validation Rule	Error Code
CAPA	RCA must exist before ROOT*CAUSE*IDENTIFIED	CAPA*NO*RCA
CAPA	At least one action before start execution	CAPA*NO*ACTIONS
CAPA	All actions verified before complete execution	CAPA*ACTIONS*NOT_VERIFIED
CAPA	Closure rules (all actions verified, effectiveness checked)	CAPA*CLOSE**
Deviation	Investigation must exist before IMPACT_ASSESSMENT	DEV*NO*INVESTIGATION
Deviation	Impact assessment must exist before DISPOSITION	DEV*NO*IMPACT_ASSESSMENT
Change Control	Impact assessment must exist before QA_REVIEW	CC*NO*IMPACT_ASSESSMENT
Change Control	All tasks completed before CLOSED	CC*CLOSE*TASKS_INCOMPLETE
Change Control	All approvals resolved before CLOSED	CC*CLOSE*APPROVALS_PENDING
Change Control	Training completed before CLOSED (if required)	CC*CLOSE*TRAINING_INCOMPLETE
NC	stage_detected must be valid or null	DB constraint

End of Document

Generated: 27-Jun-2026 | QMS-Pharma v1.0